

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9648/03

Paper 3 Applications Paper

17 September 2015

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
1	15
2	15
3	10
4	12
5	20
Total	72

This document consists of ** printed pages.



Section A
Answer **all** questions.

- 1 *Agrobacterium tumefaciens* is a soil bacterium commonly used in the genetic engineering of plants.

Fig. 1.1 shows the stages in the transfer of the *cry* gene from *Bacillus thuringiensis* into plant cells.

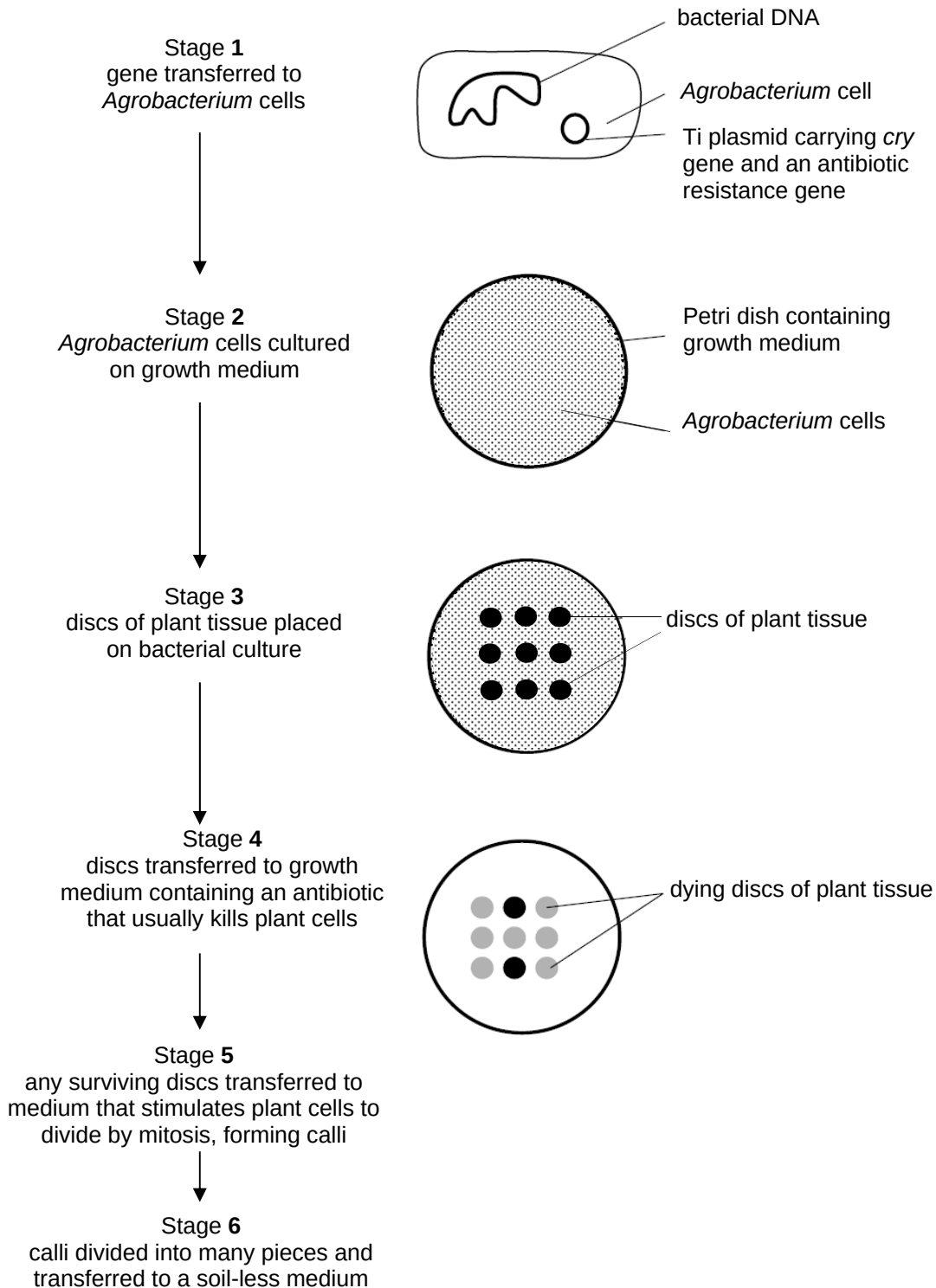


Fig. 1.1

- (a) The cry gene can be obtained by screening DNA libraries.

With reference to the differences between genomic and cDNA libraries, explain which DNA library is preferred for cloning of the cry gene into plant cells.

1. cDNA library containing cDNA that has been reverse transcribed from mRNA;;

2. cDNA library allows intact gene to be obtained vs genomic library where gene may be cut in between if there is a RS within the seq;;

® cDNA lacks introns

[2]

- (b) Describe how the Ti plasmid containing the cry gene is produced.

1. restriction digestion of Ti plasmid and cry gene with same restriction enz to product sticky ends;;

2. mix fragments to allow annealing via H bonding btw comp bp, followed by ligation using DNA ligase to catalyse formation of phosphodiester bond btw fragments;

[2]

- (c) Describe how the Ti plasmid is transferred into *Agrobacterium* cells.

1. heat shock transformation, where recomb DNA with CaCl_2 is incubated on ice;;

2. placed in water bath at 42C for 1-2min to allow bacteria to take up DNA;;

[2]

- (d) With reference to Fig. 1.1,

- (i) Explain the significance of the antibiotic resistance gene on the Ti plasmid.

1. acts as selectable marker, conferring resistance against antibiotic (e.g. coding for enz to break down antibiotics) to allow selection of plant cells containing plasmid;

2. in stage 4, plant cells that did not uptake plasmid containing antibiotic resistance gene die when placed in medium containing antibiotic;

[2]

- (ii) Suggest the relative concentrations of plant growth regulators present in the nutrient medium in Stage 5.

intermediate ratios of auxin:cytokinin;

[1]

The fall armyworm is a common insect pest of field corn. The use of spray insecticides is ineffective as the larvae are protected once they enter the stem.

Two types of Bt corn, YieldGard and Herculex, have been genetically modified with variants of the *cry* gene coding for different forms of Bt toxin.

The performances of YieldGard and Herculex technologies for their effects on fall armyworm survival and corn damage were compared with a conventional non-Bt corn line. The results of the study are shown in Fig. 1.2.

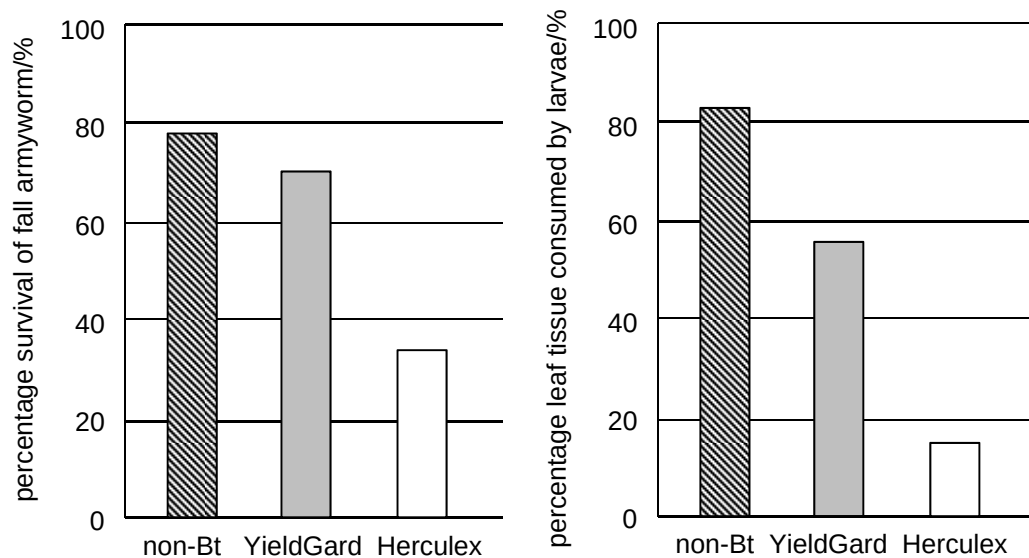


Fig. 1.2

(e) With reference to Fig. 1.2,

(i) Describe the effect of Bt toxin on the fall armyworm.

1. **fall armyworm survival decreases in plants producing Bt toxin, with 78% survival in non-Bt plants, and 70% and 35% in YieldGard and Herculex plants;;**
2. **fall armyworm that ingest Bt toxin are not able to survive due to intestinal lysis;;**

[2]

(ii) Deduce which type of Bt corn will best increase crop yield.

1. **Herculex;;**
2. **with 16% leaf tissue consumed compared to 82% in non-Bt and 58% in YieldGard;;**

[2]

(f) Describe **two** concerns of genetically modified Bt corn.

1. **evolution of resistance in organisms as Bt toxin can act as selection pressure selecting for resistant individuals in the population;;**
2. **undesirable environmental impact due to leaching into the soil;;**

3. *unintended effects on other organisms, e.g. Monarch butterfly killed by milkweed contaminated by pollen from Bt corn;;*
4. *possibility of gene flow, where transgene in Bt corn may be transferred through cross-pollination to wild-type relatives;;* [2]

[Total:15]

2 A cybrid (cytoplasmic hybrid cell) is produced as shown in Fig. 2.1.

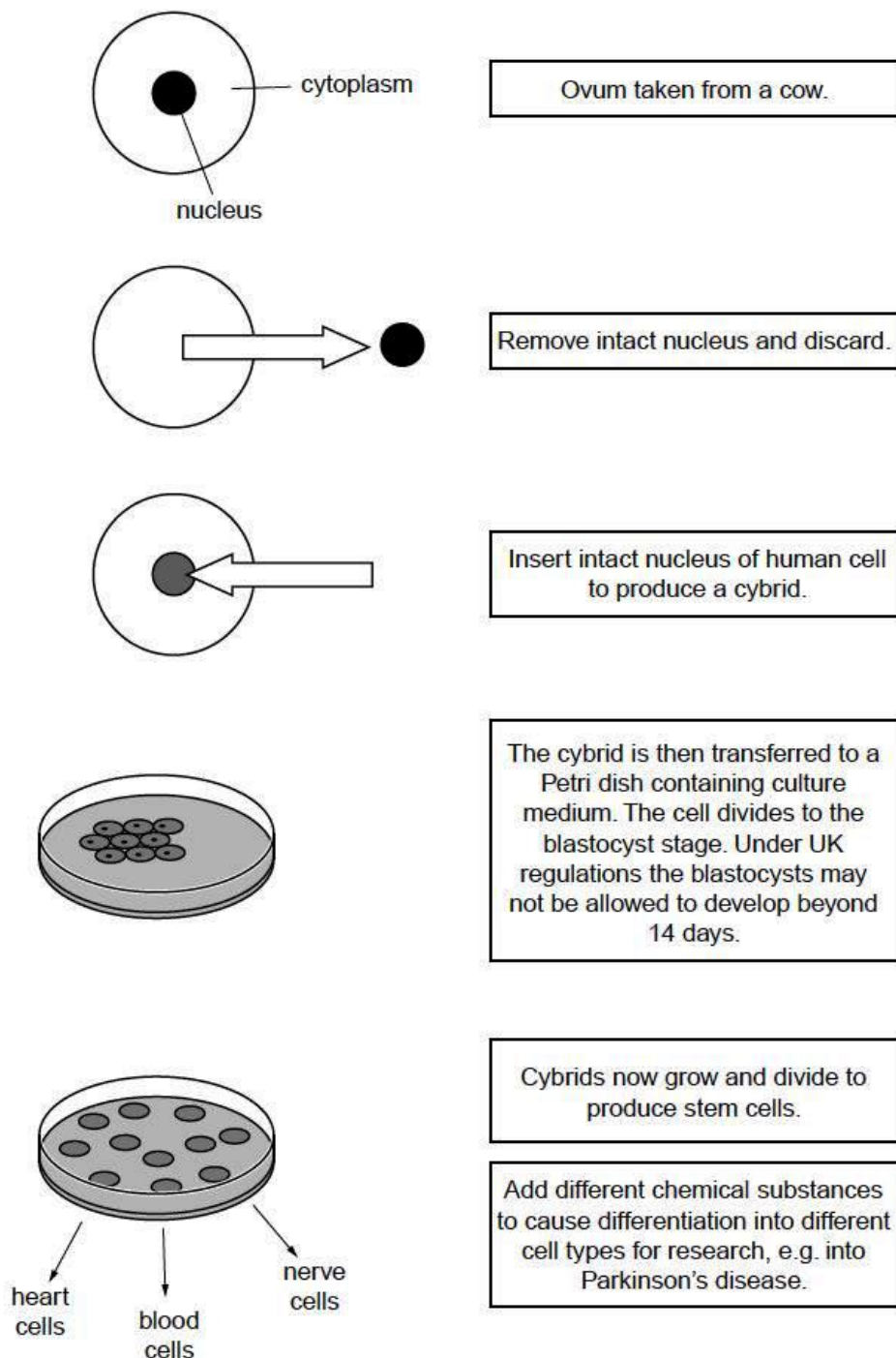


Fig. 2.1

- (a) The DNA of such a cybrid is 99.6% human. The remaining 0.4% of the DNA is from cow in the cytoplasm. Explain why there is DNA in the cytoplasm of the cybrid.

1. presence of DNA in mitochondria;;

2. which is contributed by the ovum to the zygote;;

[2]

- (b) Suggest why UK regulations do not allow blastocysts to develop beyond 14 days.

1. blastocysts beyond 14 days has good chance of surviving as an embryo (thus considered a 'life');;

[1]

- (c) (i) State which cells are taken from the blastocyst to produce stem cells.

1. inner cell mass;;

[1]

- (ii) Describe the features of these stem cells.

1. pluripotent

capable of differentiating into any cell type in the adult except placenta;;

2. unlimited self-renewal capacity

able to divide unlimited no. of times due to active telomerase / ⊗ undergo replicative senescence;;

3. undifferentiated / unspecialised

no tissue specific structures;;

[3]

- (d) (i) Suggest how addition of different chemical substances can cause stem cells to differentiate into different cell types.

1. chemical signals e.g. hormones can bind cell surface receptors

to trigger cell signalling pathway / cause signal transduction;;

2. to switch on / activate specific groups of genes

coding for specific cellular struct e.g. axons in neurones;;

[2]

- (ii) Explain if the heart cells, blood cells and nerve cells formed are human or bovine (cow).

1. human

tissue specific structures are coded by alleles / genes in chromosomes in nucleus which is of human origin;;

[1]

- (iii) Suggest how the resultant cells can be used for research, e.g. Parkinson's disease.

1. drug testing;;

2. expose neurones to drug under test to observe regeneration / progress of degeneration;;

[2]

- (e) When the Human Fertilisation and Embryology Bill was considered by the UK Parliament in 2008, some people argued that it is unethical to allow the production of cybrids.

State whether you **agree** or **disagree** that this is unethical and explain why you reached this decision.

1. state stand;;

2. [agree] presence of human and animal DNA means it is partly human and partly cow / a human animal hybrid which would not happen in nature;;

3. claims of the benefits of embryonic stem cell research are over-rated / few (if any) examples of success in medical applications;;

4. dangerous precedent / thin end of the wedge argument / current applications (may be benign) but will lead to abuse in future;;

5. unnecessary because, there are alternative sources of stem cells e.g. use of adult stem cells/umbilical cord cells;;

6. [disagree] the amount of non-human DNA is very small/negligible ;

7. protocol limits keeping blastocyst to 14 days of development thus cannot develop into a whole organism ;

8. ovum has bovine proteins which would be rejected at implantation;;

9. provides a more ethical alternative to the use of embryonic stem cells from humans;;

10. can provide many more stem cells than is possible from embryonic stem cells;;

11. it would be unethical not to use cybrids to relieve, human suffering / e.g. diseases such as Parkinson's disease;;

[3]

[Total: 15]

- 3 Cystic fibrosis affects epithelial cells lining organs such as the lungs and intestines leading to accumulation of mucus. In 1989, the mutated gene responsible for cystic fibrosis was identified. This opened the way for the development of gene therapy to reduce the symptoms of the disease. The idea has been to introduce a normal copy of the gene into the cells of patients.

- (a) (i) Describe **one** mutation that leads to CF.

1. $\Delta F508$ / 3 base deletion at codon 508;;

2. resulting in deletion of phe (missense mutation) in CFTR prot;;

[2]

- (ii) Explain how the mutation stated in (a)(i) results in mucus accumulation in lung epithelial.

1. cause Δ in 3D config of CFTR prot (at NBD)

leading to inability to trspt Cl⁻ out of epithelial cells;;

2. Na⁺ is retained to balance electrical charges

resulting in lower water potential within the cell;;

3. retains water / water does not move out of cell by osmosis

causing viscous mucus which cilia cannot effectively remove;;

[3]

Many attempts have been made to find methods of using gene therapy to treat cystic fibrosis. One approach uses viruses to deliver normal alleles of the CFTR gene into epithelial cells of the airways. Viral delivery systems have two main problems:

- The virus may trigger an immune response which destroys the infected cells.
- Most non-pathogenic viruses are not very good at getting into cells, so very few cells receive the allele.

A team of researchers in the USA developed a new strain of adeno-associated virus, AAV2.5T. Genes for the CFTR protein and for an enzyme, luciferase, were added to the DNA of the viruses. Luciferase produces a fluorescent green protein when luciferin is added.

The normal AAV strain and the AAV2.5T strain were added to cultures of epithelial cells from the airways. After adding luciferin, the numbers of cells that had taken up the CFTR gene was estimated using the intensity of the green fluorescence which developed.

The results are shown in Fig. 3.1

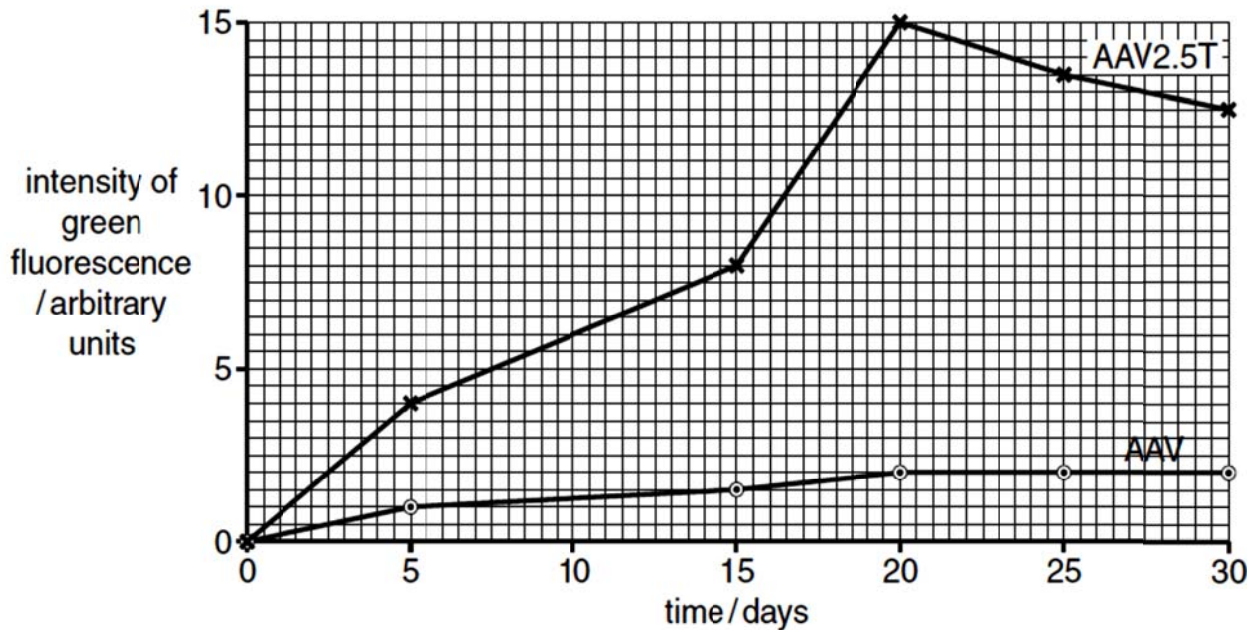


Fig. 3.1

- (b) Explain why the intensity of green fluorescence can be used to estimate the number of cells that have taken up the CFTR gene.

1. as gene coding for luciferase is intro together with CFTR gene into viral DNA

epithelial cells which have taken up CFTR gene would also have the luciferase gene;;

2. thus successfully transfected epithelial cells would express the luciferase enz

that converts luciferin into a fluorescent green prot;;

[2]

- (c) With reference to Fig. 3.1, explain the difference in ability of the two viral strains, AAV and AAV2.5T, to infect epithelial cells from the airways.

1. AAV2.5T strain is able to infect epithelial cells more effectively

quote values;;

2. AAV2.5T has modified surface receptors

that bind specifically to surface prot of lung epithelial cells;;

[2]

- (d) State **one** other advantage of using AAV to deliver genes into cells.

1. integrates transgene into specific location on chr 19

allowing stable propagation to daughter cells;;

OR unlikely to cause insertional mutagenesis;;

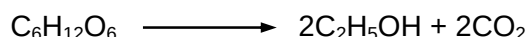
[1]

[Total: 10]

4 Planning question

Yeast is a single cell eukaryotic fungus used in industry for the production of drinks containing ethanol. The yeast is grown with a carbohydrate energy source in anaerobic conditions. During this fermentation process the yeast oxidises the carbohydrate to synthesise ATP for growth and produces ethanol and carbon dioxide as by-products.

The equation for anaerobic respiration is:



You are required to find out the rate of respiration of yeast using different carbohydrates.

You are provided with the following materials and apparatus which you must use:

- activated yeast culture
- 10% fructose
- 10% glucose
- 10% lactose
- 10ml syringes
- boiling tubes
- thermostatically regulated water bath and thermometer

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated
- pipettes, glass rods etc.
- stopwatch

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram(s), if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

Theoretical b/grd [2m]

- 1. ref to glc as primary resp substrate that enters glycolysis & subsequently ethanol in anaerobic resp w release of CO₂;;***
- 2. ref to diff carbohydrate sources e.g. disaccharides lactose also enter resp rxn but at diff points thus to different efficiency;;***

Independent Variable [1m]

- 3. fixed vol of type of respiratory substrate / carbohydrate;;***

Dependent Variable [1m]

- 4. rate of respiration measured by rate of CO₂ production with feasible means of gas collection e.g. gas syringe, displacement etc.;;***

Controlled Variable [2m]

- 5. temperature at 37°C maintained by thermostatically regulated waterbath;;***
- 6. state constant vol of yeast to keep no. of cells approximately constant;;***

Control Setup

- 7. describe control setup with boiled yeast;;***

Procedure [1m] (any 2 points)

8. *equilibration for stated time e.g. 2min;*

9. *stir for homogenous yeast suspension;*

10. *state reasonable reaction duration;*

Reliability [1m]

11. *3 replicates, 2 repeats;;*

Results [1m]

12. *table of results & theoretical bar graph;;*

Risk Assessment [1m]

13. *[hazard] yeast, [risk] skin irritant, [safety] wear latex gloves;;*

[illegible]

This image shows a full page of white paper designed for handwriting practice. It features 20 evenly spaced, horizontal dashed lines that run across the entire width of the page. There are no margins, text, or other markings present.

Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)**, etc., as indicated in the question.

5 (a) Discuss the problems associated with cloning of eukaryotic genes using *E. coli* cells. [5]

- 1. eukaryotic genes contain introns and exons;;**
- 2. pre-mRNA transcribed require post-transcriptional modification to produce mature mRNA, where splicing occurs to remove introns and join exons, catalysed by spliceosomes;;**
- 3. prokaryotic host unable to perform post-transcriptional mod, thus mRNA translated contain introns → non-functional prot;;**
- 4. proteins of eukaryotic genes may require post-translational mod;;**
- 5. e.g. addition / deletion / substitution of oligosaccharide chains which occurs in Golgi body;;**
- 6. prokaryotic host do not contain memb-bound organelles, therefore unable to modify polypeptide to produce functional prot;;**

(b) Explain how problems associated with cloning of eukaryotic genes using *E. coli* cells are overcome. [6]

- 1. use mature mRNA as starting material (obtained via column chromatography), performing reverse transcription of ss mRNA to ds cDNA;;**
- 2. syn DNA strand by adding free dNTPs comp to mRNA by reverse transcriptase forming RNA-DNA hybrid (using oligo d(T) sequences as primer);;**
- 3. degradation of RNA strand by RNase to obtain ssDNA; ;**
- 4. formation of hairpin loop (which acts as primer) by RTase, DNA pol synthesises second strand using first cDNA strand as template / terminal transferase adds multiple C residues to 3' end, addition of oligo d(G) primer to bind C residues;;**
- 5. syn DNA strand by adding free dNTPs comp to 1st DNA strand by DNA pol to form complementary ds cDNA;;**
- 6. mRNA transcribed from cDNA does not contain introns, which does not require splicing;;**
- 7. chemical syn / addition of enzymes;;**
- 8. for post-translational mod of polypeptide to yield functional prot;;**

(c) Rubisco is an important enzyme in the Calvin cycle.

Outline how it might be possible to produce genetically modified plants with increased rubisco concentrations, or rubisco with enhanced efficiency. [9]

- 1. obtain rubisco gene coding for rubisco of higher efficiency from another**

organism (e.g. bacteria) / **design construct consisting strong promoter and rubisco gene;**

2. restriction digestion of gene / construct with RE (e.g. EcoRI) and restriction digestion of plasmid vector using same RE to produce sticky ends;;
3. allow gene of interest to anneal with vector via hydrogen bonds between comp bp;
4. ligation involving formation of phosphodiester bond between DNA fragments;
5. heat shock transformation of *Escherichia coli* / *Agrobacterium tumefaciens* to allow bact to take up DNA by incubating mixture of DNA and CaCl_2 in ice bath, followed by 42°C for 1-2min;
6. select for bact that has taken up recomb plasmid (e.g. via replica plating);
7. isolate recomb plasmid from *E.coli* and coat tungsten beads with DNA and fire at high speed to plant cells in callus culture via particle bombardment / allow *A. tumefaciens* to infect plant cells;
8. screen for plant cells that have taken up recomb plasmid;
9. culture plant cells by placing in ideal growth conditions, such as optimum light intensity, temp, and humidity, and nutrient medium (e.g. containing sucrose, aa, vitamins);
10. alter plant growth regulators, where high auxin to cytokinin ratio promotes root growth and high cytokinin to auxin ratio promotes shoot growth, to allow it to grow into a plantlet;
11. transplant plantlet for hardening and acclimatization and subsequently to soil to grow into a whole plant;
12. genetically modified plant will contain gene coding for rubisco of higher efficiency / **construct consisting strong promoter and rubisco gene;**
13. resulting in production of rubisco with higher catalytic activity / **resulting in inc transcription of rubisco gene and inc production of rubisco;**

[Total: 20]

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